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Section - Cyber

Assignment - 1
(Course: Fundamental of Cyber Security (TCS 495))

Ques-1 Explain the importance of Cyber Security in today's digital World.
Give five real-life examples where cyber security is necessary.

Ans Cyber Security is crucial in today's digital World due to the increasing reliance on technology and the internet. It protects sensitive data from unauthorized access, ensures the integrity of information and maintains the Availability of services. The importance of Cyber Security can be summarized as follows.

- Protection of Personal and financial information.
- Prevention of Cyber Attacks that can disrupt services.
- Safeguarding organizational reputation and trust.
- Compliance with legal and regulatory requirements.
- Mitigation of risks associated with data breaches.

Five real-life examples of where Cyber Security is necessary:

- Online banking system to protect user financial data.
- E-commerce platforms to secure customer transactions.
- Healthcare systems to safeguard patient records.
- Government databases to ~~protect~~ prevent unauthorized access to sensitive information.
- Corporate networks to protect intellectual property and trade secrets.

Ques-2 Compare the following types in tabular form:

- White Hat
- Grey Hat
- Black Hat

Also mention one real-world scenario for each

Hacker's Type	Description	Real-World Scenario
White Hat	Ethical hacker who helps organizations secure their systems.	A security consultant conducting penetration testing for a bank.
Grey Hat	Hackers who may violate laws but do not have malicious intent.	A hacker exposing vulnerabilities in a website without permission.
Black Hat	Malicious hackers who exploit system for personal gain.	A hacker stealing credit card information from an e-commerce site.

Ques-3 Explain the Zero trust Model. Design a security strategy for a college computer using Zero Trust principles.

Ans The Zero trust Model is a security framework that assumes that threats can be external and internal. It operates on the principle of never trust, always verify meaning that every access request must be authenticated and authorized regardless of its origin.

A security strategy for a college computer lab using zero trust:

- Implement multi-factor Authentication for all users.
- Segmenting the network to limit access to sensitive resources.
- Regularly monitoring and logging user activity.
- Enforcing strict access controls based on user roles.
- Conducting regular security assessments and updates.

Explains CIA Triad: Confidentiality, Integrity, Availability.
Give examples of each Pillar.

Ans

The CIA Triad is a model designed to guide policies for information security. It consists of three core principles.

- Confidentiality: Ensures that sensitive information is accessed only by authorized users.

examples - A data breach where personal information is leaked.

- Integrity: ~~At~~ Ensures that data is accurate and unaltered.

examples - A ransomware attack that modifies files.

- Availability: Ensures that information and resources are accessible when needed.

examples - A Distributed Denial of Service (DDoS) attack that disrupts service access.

Ques 5 A Company's employee accounts were accessed by an unknown user at midnight. Files were modified and logs deleted. Answer the following.

- Identify possible cyber threats involved.
- What should be analyzed first?
- Suggest treatment steps.
- Which CIA Pillars were affected?

Ans

- Possible cyber threats involved include:

- Unauthorized access (hacking)
- Data manipulation (file modification)
- Log tampering (deletion of logs)

- The first thing to analyze should be the access logs to determine how the attacker gained entry.

- c) Suggested treatment steps include:
- Isolating affected systems to prevent further damage.
 - Restoring files from backups if available.
 - Conducting a forensic analysis to identify the breach method.
 - Implementing stronger access controls and monitoring.

- d) The affected CIA Pillars are -
- Confidentiality (due to unauthorized access)
 - Integrity (due to file modifications)
 - Availability (potential disruption of services).

Ques-6

A student installs cracked software and later all files become encrypted.

- What will happen?
- How did it occur?
- Prevention methods.
- Recovery strategy.

Ans

- The attack that happened is a ransomware attack.
- It occurred when the student installed the cracked software which likely contained malicious code that encrypted files.
- Prevention methods include:
 - Avoiding the installation of cracked software.
 - Using reputable antivirus software.
 - Regularly updating software and operating systems.
- Recovery strategy includes:
 - Restoring files from backups if available.
 - Using decryption tools if the ransomware is known.
 - Reinstalling the operating system if necessary.

- 7 User Mangment : Escail commands for :
- Create user student 1
 - Set Password.
 - Add user to sudo group.
 - Switch to that user.
 - Delete the user.

Write commands with explanation.

Ans

sudo adduser student1

Creates a new user named student1

sudo passwd student1

Sets a password for student1

sudo usermod -aG sudo student1

Adds student1 to the sudo group.

su - student1

Switches to the student1 user.

sudo deluser student1

Deletes the user student1

```
(kali@kali)-[~/Assignment1]
$ sudo adduser Naitik
[sudo] password for kali:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for Naitik
Enter the new value, or press ENTER for the default
Full Name []:
Room Number []:
Work Phone []:
Home Phone []:
Other []:
Is the information correct? [Y/n] y

(kali@kali)-[~/Assignment1]
$ sudo useradd Naitik
useradd: user 'Naitik' already exists

(kali@kali)-[~/Assignment1]
$ sudo passwd Naitik
New password:
Retype new password:
passwd: password updated successfully

(kali@kali)-[~/Assignment1]
$ sudo usermod -aG sudo Naitik

(kali@kali)-[~/Assignment1]
$ su - Naitik
Password:
(Naitik@kali)-[~]
$ sudo userdel -r Naitik
[sudo] password for Naitik:
userdel: user Naitik is currently used by process 3987
```

Ques-8

File Operations: → Create a folder Assignment1 and perform

- Create file notes.txt.
- Write text into file.
- Append new text.
- Copy file.
- Move file.
- Delete copied file.
- Display permissions.
- Change permissions to read/write only for owner.

Ans.

mkdir Assignment1

Create a folder named Assignment1

cd Assignment1

Navigate into the Assignment1 folder.

touch notes.txt.

Create a file named notes.txt.

echo "Initial test" > notes.txt.

Write initial text into notes.txt.

echo "Additional text" >> notes.txt.

Appends additional text to notes.txt.

cp notes.txt notes_copy.txt.

Copies notes.txt to notes_copy.txt.

mv notes_copy.txt ../

Moves notes_copy.txt to parent directory.

test -f notes_copy.txt && rm notes_copy.txt

Deletes the copied file if it exists.

ls -l notes.txt.

Displays permissions of notes.txt.

chmod 600 notes.txt.

Changes permissions to read/write only for the owner.


```

(kali@kali)-[~]
└─$ mkdir Assignment1
(kali@kali)-[~]
└─$ cd Assignment1
(kali@kali)-[~/Assignment1]
└─$ touch notes.txt
(kali@kali)-[~/Assignment1]
└─$ echo "This is my assignment file." > notes.txt
(kali@kali)-[~/Assignment1]
└─$ echo "Adding more content" >> notes.txt
(kali@kali)-[~/Assignment1]
└─$ cp notes.txt notes_copy.txt
(kali@kali)-[~/Assignment1]
└─$ mv notes_copy.txt moved_notes.txt
(kali@kali)-[~/Assignment1]
└─$ rm moved_notes.txt
(kali@kali)-[~/Assignment1]
└─$ ls -l
total 4
-rw-rw-r-- 1 kali kali 48 Apr 29 16:50 notes.txt
(kali@kali)-[~/Assignment1]
└─$ chmod 600 notes.txt

```

Ques-9

Write a bash script that:

Accepts student marks as input.

Prints grade using conditions:

90+ = A

75+ = B

50+ = C

below 50 = Fail,

Ans-

Code:-

```
#!/bin/bash
```

```
read -p "Enter student marks:" marks
```

```
if [ $marks -ge 90 ]; then
```

```
    echo "Grade: A"
```

```
elif [ $marks -ge 75 ]; then
```

```
    echo "Grade: B"
```

```
elif [ $marks -ge 50 ]; then
```

```
    echo "Grade: C"
```

```
else
```

```
    echo "Grade: Fail"
```

```
fi
```

```

(kali@kali)-[~/Assignment1]
└─$ nano script1.sh
(kali@kali)-[~/Assignment1]
└─$ chmod +x script1.sh
chmod: cannot access 'script1.sh': No such file or directory
(kali@kali)-[~/Assignment1]
└─$ chmod +x script1.sh
(kali@kali)-[~/Assignment1]
└─$ ./script1.sh
Enter marks:
98
Grade: A
(kali@kali)-[~/Assignment1]
└─$ ./script1.sh
Enter marks:
67
Grade: C
(kali@kali)-[~/Assignment1]
└─$

```

Ques-10

Write a Bash script using loop to:
Create 10 files named file1.txt to file10.txt.
Write current date in each file.

Ans

```
#!/bin/bash
for i in {1..10}; do
    echo $(date) > file$i.txt
done
echo "10 files created successfully"
```

```
(kali@kali) [~/Assignment1]
$ nano script2.sh
(kali@kali) [~/Assignment1]
$ chmod +x script2.sh
(kali@kali) [~/Assignment1]
$ ./script2.sh
10 files created successfully
(kali@kali) [~/Assignment1]
$ ls
file10.txt file1.txt file2.txt file3.txt file4.txt file5.txt file6.txt file7.txt file8.txt file9.txt
(kali@kali) [~/Assignment1]
$
```

Ques-11

Write a Python program to create a dictionary containing 5 students names and.
Print topper name and average marks.

Ans

```
students = {"Alice": 85, "Bob": 90, "Charlie": 78, "David": 88, "Eve": 92}
topper = max(students, key=students.get)
Average_marks = sum(students.values()) / len(students)
print("Topper:", topper)
print("Average Marks:", average_marks)
```

```
(kali@kali) [~/Assignment1]
$ nano script3.py
(kali@kali) [~/Assignment1]
$ python3 script3.py
Topper: Simran
Average Marks: 87.6
```


12. Write a Python code to take a sentence as input and counts:

Total words.

Total vowels

Total consonants.

Frequency of each word.

Ans Code:

```
sentence = input("Enter a sentence: ")
words = sentence.split()
word_count = len(words)
vowel_count = sum(1 for char in sentence if char.lower() in 'aeiou')
consonant_count = sum(1 for char in sentence if char.isalpha() and char.lower() not in 'aeiou')
word_frequency = {word: words.count(word) for word in words}
print("Total words:", word_count)
print("Total vowels:", vowel_count)
print("Total consonants:", consonant_count)
print("Word frequency:", word_frequency)
```

```
(kali@kali)-[~/Assignment1]
$ nano script4.py

(kali@kali)-[~/Assignment1]
$ python3 script4.py
Enter a sentence: Hello World!
Total Words: 2
Total Vowels: 3
Total Consonants: 7
Word Frequency: {'Hello': 1, 'World!': 1}
```

Bonus Challenge

Create a new project: Cyber Safety Tool:
→ features =

- Password strength checker.
- File backup tool
- Log analyzer.

Code

Cyber Safety Toolkit

import os, shutil.

from datetime import datetime

def toolkit():

print("In --- Cyber Safety Toolkit ---")

print("1. Password checker | 2. File Backup | 3. Log Analyzer | 4. Exit |")

choice = input("Select an option")

if choice == '1':

pwd = input("Enter Password: ")

score = sum([len(pwd) >= 8, any(c.isdigit() for c in pwd),
any(c.isupper() for c in pwd), any(not c.isalnum()
for c in pwd)])

print(f"Strength: {['Very Weak', 'Weak', 'Medium', 'Strong',
'Very Strong'][score]}")

elif choice == '2':

file = input("File name to backup: ")

if os.path.exists(file):

dest = f"{file}_ {datetime.now().strftime('%H%M%S')
}.bak"

shutil.copy(file, dest)

print(f"Backup created: {dest}")

else: print("File not Found.")

elif choice == '3':

log = ["Failed password for root", "Accepted password for user",
Failed for admin"]

failed = sum(1 for line in logs if "Failed" in line)

elif choice == '4': exit()

tool kit()

tool kit()

```
nsaitikdhiman@nsaitiks-Laptop prac % nano script.py
nsaitikdhiman@nsaitiks-Laptop prac % python3 script.py

--- Cyber Safety Toolkit ---
1. Password Checker
2. File Backup
3. Log Analyzer
4. Exit
Select an option: 1
Enter password: Iran@man94
Strength: Very Strong

--- Cyber Safety Toolkit ---
1. Password Checker
2. File Backup
3. Log Analyzer
4. Exit
Select an option: 2
Filename to backup: search.cpp
Backup created: search.cpp_170848.bak

--- Cyber Safety Toolkit ---
1. Password Checker
2. File Backup
3. Log Analyzer
4. Exit
Select an option: 3
Analysis: 2 failed login attempts detected.

--- Cyber Safety Toolkit ---
1. Password Checker
2. File Backup
3. Log Analyzer
4. Exit
Select an option: 1s

--- Cyber Safety Toolkit ---
1. Password Checker
2. File Backup
3. Log Analyzer
4. Exit
Select an option: 4
nsaitikdhiman@nsaitiks-Laptop prac % ls
a.out                                script.py                            search.cpp                            search.cpp_170848.bak
nsaitikdhiman@nsaitiks-Laptop prac %
```